TCP BASED CHATING APPLICATION

Objective:

Here the objective is to implement **chating** application to allow two **persons** for chating over a network i.e.., **CLIENT1 and CLIENT2 ,** with help of **SOCKET** programming by using **TCP** protocols.

TASK:

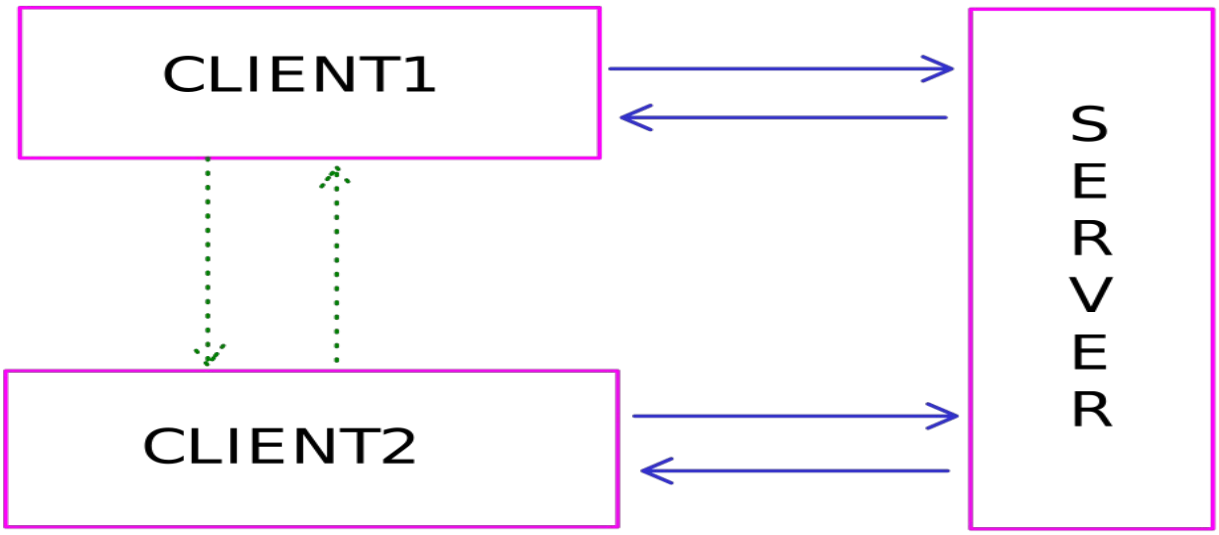
Here **client1** and **client2** can not communicate directly, So we need **server** to handle **client1** and **client2**.

There are three versions can be posible i.e , simplex , half duplex and full duplex.

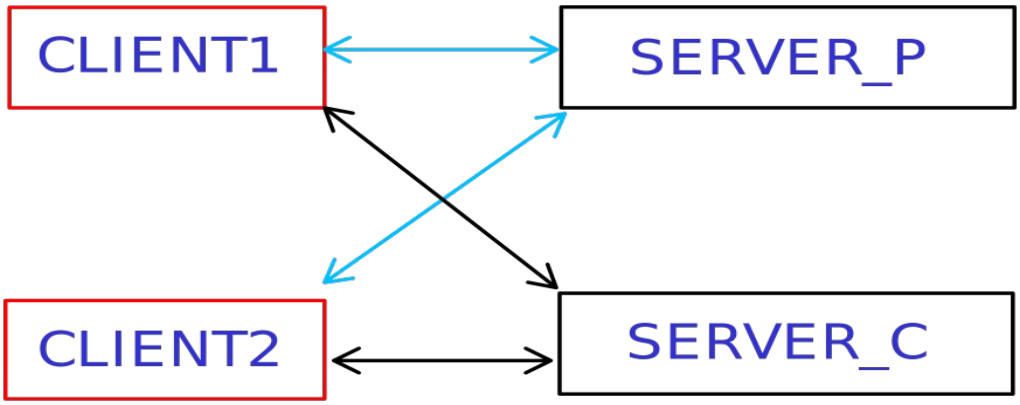
Consider **FULL DUPLEX ,** Client1 will send message to server and that server will send received message to client2 and vice versa.

Approuch:

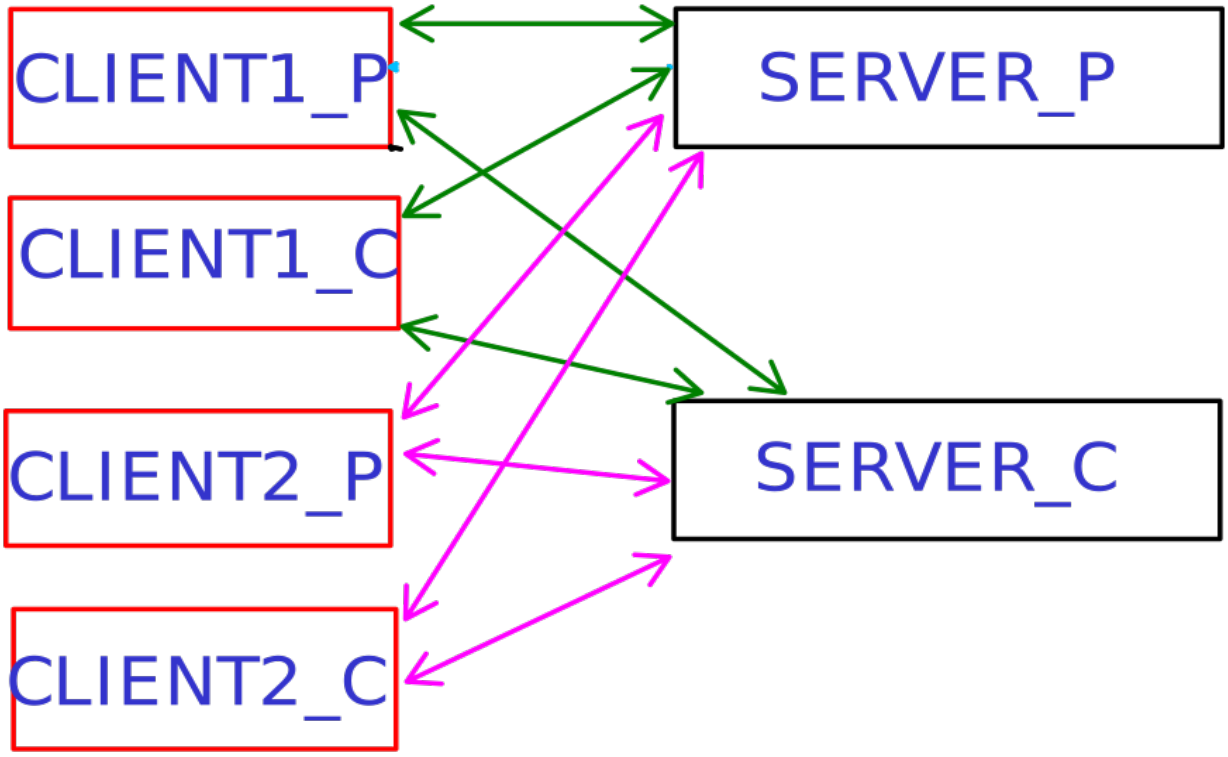
- Server should accept two clients connection with help of two accept() calls i.e.., from client1 and client2



- After connection estlabished , create child server. So , now we have two server processes i.e **parentServer** and **childServer**. Both are having connection with client1 and client2

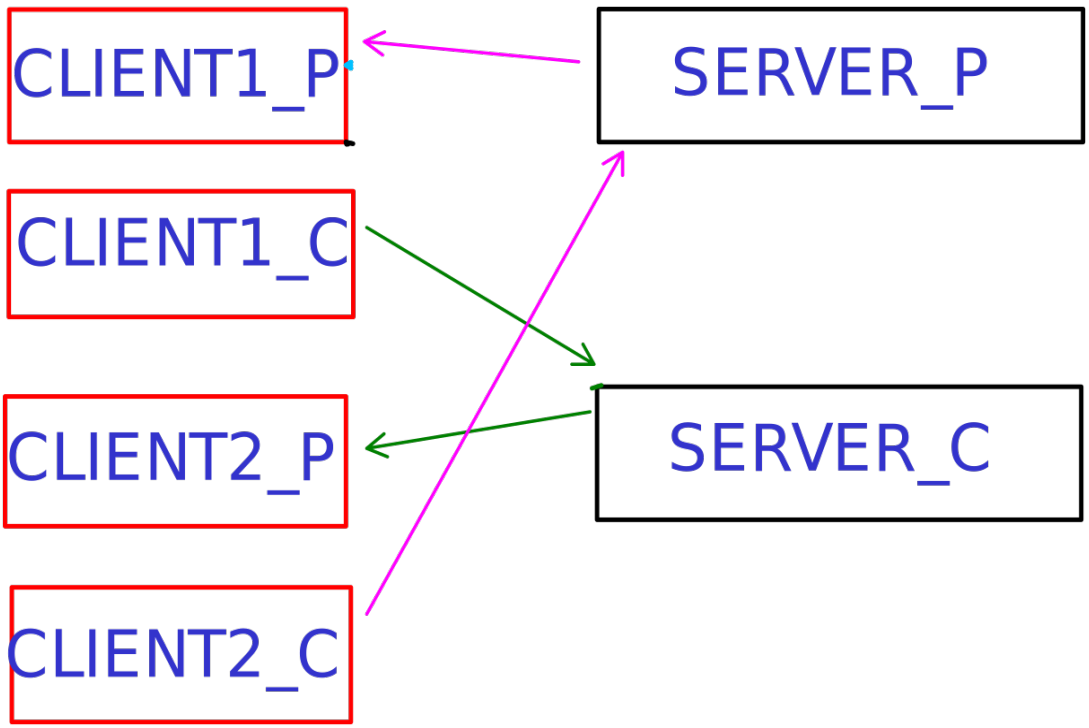


- Now our task is to make full duplex, So , we need to create child process to clients , after connect() call in clients.



- Now to make full duplex, manage your **if** and **else** blocks in clients and server as follows.
    - **client1_child** always send MSG to **server_child**
    - **server_child** always first recv MSG from **client1_child** and send to **client2_parent**.
    - **client2_parent** always recv MSG from **server_child**

    - **client2_child** always send MSG to **serever_parent**.
    - **server_parent** always first recv MSG from **client2_child** and send to **client1_parent**.
    - **client1_parent** always recv MSG from **server_parent**



THANK YOU